

MD ASHIKUR RAHMAN

Lead AI Engineer | Multimodal AI, Generative AI, and Production Computer Vision

Dhaka, Bangladesh | Open to Relocation and Remote Opportunities

mdashikur.rafi@gmail.com | +880 1675 964 080 | [Portfolio](#) | [GitHub](#) | [LinkedIn](#) | [Google Scholar](#)

Summary

Lead AI Engineer specializing in multimodal AI, vision-language systems, generative AI, and production computer vision. Lead 15+ engineers and researchers across applied research, product development, and enterprise delivery. Built and scaled production AI systems supporting 200+ brands, including computer-vision platforms that have processed 4.5M+ images globally.

Experience

Lead AI Engineer, The KOW Company

Jan 2023-Present

- Lead a multidisciplinary team of 15+ ML engineers, software engineers, and researchers across architecture, planning, code review, and delivery; maintain 90-95% on-time delivery across AI engineering projects.
- Oversee development of production AI systems across virtual try-on, generative media, 3D reconstruction, catalog audit, and audio QA.
- Direct applied research on AI hallucination, prompt safety, visual grounding, and video understanding.
- Develop and scale production segmentation and image-processing systems, including Retouched.ai; publish reproducible research code, datasets, and model artifacts on GitHub and Hugging Face.

Senior Machine Learning Engineer, The KOW Company

Jul 2021-Dec 2022

- Improved object detection and segmentation performance by 20-35% across internal evaluation benchmarks; the resulting models were later deployed through Retouched.ai.
- Led 6+ client ML engagements from business requirements to technical delivery; built offline evaluation pipelines and production A/B testing workflows to validate model quality, inference performance, and production outcomes.

Machine Learning Engineer, The KOW Company

Jul 2020-Jun 2021

- Built deep learning models for production object recognition, image segmentation, and background-removal workflows; developed scalable preprocessing, training, and A/B testing pipelines.

Senior Software Engineer, Smart Technologies (BD) Ltd

Sep 2016-Dec 2019

- Led .NET and SQL Server supply-chain systems on a 4 TB database, reducing report generation from 20 minutes to 40-54 seconds.
- Achieved 70-75% process automation and 99.9% synchronization success for offline-capable enterprise workflows.

Software Engineer, Proggasoft

Mar 2015-Aug 2016

- Developed ASP.NET MVC features, backend services, and database integrations for DevSkill.com.

Key Projects

Retouched.ai - Object Detection and Segmentation

Production

- Developed production salient-object segmentation for background removal; improved segmentation quality by 17% on internal benchmarks, reduced processing time by 30%, supported uploads up to 257 MB, and achieved a 2.27-second average processing time across standard production workloads.
- Scaled Retouched.ai to process 4.5M+ images globally for hundreds of customers using PyTorch, U²-Net-inspired salient-object segmentation, FastAPI, and Google Cloud Platform.

Omnimage.ai - AI Image and Video Generation

Production

- Co-designed and launched production image- and video-generation APIs used across 200+ brands for creative and product-image workflows.
- Built workflows for reference-image conditioning, asynchronous processing, prompt classification, intent routing, and automated model selection.

The Fitting Room - Cross-Brand Virtual Try-On Platform

In Development

- Conceived and co-led a unified cross-brand virtual try-on platform covering 170+ brands.
- Architected a Dockerized FastAPI/Nginx backend using SQL Server, Google Cloud Storage, Redis, recommendation services, and 2D virtual try-on pipelines.

Enterprise AI Catalog Audit and Image Quality Assurance Platform

In Development

- Lead development of an AI catalog-audit and image quality assurance platform for a major US retail client.
- Automate image QA, catalog validation, metadata checks, and content-health monitoring across DAM and CMS workflows using computer vision, NLP, PyTorch, and FastAPI.

CogniX - Proprietary Multimodal Image Generation and Editing

Active R&D

- Leading early-stage R&D for a proprietary multimodal image-generation and editing system for product-visualization workflows.
- Researching and prototyping multi-reference fusion and prompt-guided image-editing workflows for garment replacement, scene composition, product visualization, and style transfer using PyTorch, Diffusers, ComfyUI, and parameter-efficient fine-tuning (LoRA, QLoRA).

Peer-Reviewed Publications

- Stroke-Level Connectivity Verification: Grounding Vision-Language Models Against Topology Hallucination in Diagram Understanding. Accepted for oral presentation at ICDAR 2026; Corresponding Author.
- Automated Detection of Diabetic Retinopathy using Deep Residual Learning. IJCA, 2020.

Preprints and Manuscripts Under Review

- Step-Level Visual Grounding Faithfulness Predicts Out-of-Distribution Generalization in Long-Horizon Vision-Language Models. arXiv preprint, 2026. Under review. arxiv.org/pdf/2603.06828
- Beyond Dominant Patches: Spatial Credit Redistribution for Grounded Vision-Language Models. arXiv preprint, 2026. Under review. arxiv.org/pdf/2602.22469

Education

B.Sc. in Computer Science and Engineering

American International University-Bangladesh | 2011-2015

CGPA: 3.87/4.00; WES equivalent: 3.94/4.00

Magna Cum Laude; Top 3%; Merit Scholarship and Tuition Fee Waiver

Awards

- Champion, BASIS National ICT Awards 2020 (Retouched.ai); Finalist, APICTA 2021

Languages

- Bengali (Native); English (Professional working proficiency)

Skills

Multimodal and Vision-Language AI: Python, PyTorch, Vision-Language Models, Visual Grounding, Hallucination Evaluation, Multimodal Learning

Generative AI and LLMs: Diffusion Models, Large Language Models (Llama, Qwen), Prompt Safety, Parameter-Efficient Fine-Tuning (LoRA, QLoRA), Prompt-Guided Image Editing

Computer Vision: Object Detection, Segmentation, Pose Estimation, Image Quality Assurance

3D Vision: Structure from Motion, Multi-View Stereo, COLMAP, Open3D, Neural Radiance Fields (NeRF), 3D Gaussian Splatting

Production ML and MLOps: Model Serving, Production Deployment, Offline Evaluation, Production A/B Testing, Model Evaluation, Dataset Design

Backend and Cloud: FastAPI, Docker, Nginx, Redis, Google Cloud Platform, Google Cloud Storage, SQL Server